$$
\begin{cases}
\rho \dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial u}{\partial x}\right) = f & \text{in } (0,L)\times(0,T) \qquad (*) \\[2ex]
\mu(0)\dfrac{\partial u}{\partial x}(0,t) = g_0(t) & t \in (0,T] \\[2ex]
\mu(L)\dfrac{\partial u}{\partial x}(L,t) = g_L(t) & t \in (0,T] \\[2ex]
u(x,0) = u_0(x) & x \in [0,L] \\[2ex]
\dfrac{\partial u}{\partial t}(x,0) = u_1(x) & x \in [0,L]
\end{cases}
$$

$\Downarrow$ WEAK PROBLEM

$\forall t \in (0,T]$ find $u(t) \in H^1(0,L)$ s.t.

$$
\int_0^L \rho \frac{\partial^2 u}{\partial t^2} v \, dx + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} dx = \int_0^L f v \, dx + g_L(t) v(L) - g_0(t) v(0) \quad \forall v \in H^1(0,L)
$$

$\Downarrow$ FE FORMULATION

$\forall t \in (0,T]$ find $u_h(t) \in V_h$ s.t.

$$\int_0^L \rho \frac{\partial^2 u_h}{\partial t^2} v \, dx + \int_0^L \mu \frac{\partial u_h}{\partial x} \frac{\partial v_h}{\partial x} \, dx = \int_0^L f v_h \, dx + g_L(t) v(L) - g_0(t) v(0) \qquad \forall v \in V_h$$

+ INITIAL CONDITIONS

$\Downarrow$ ALGEBRAIC FORM

$$\begin{cases} M \ddot{\underline{u}}(t) + A \underline{u}(t) = \underline{b}(t) & \forall t \in (0,T] \\ \underline{u}(0) = \hat{\underline{u}}_0 \\ \dot{\underline{u}}(0) = \hat{\underline{u}}_1 \end{cases}$$

SYSTEM OF LINEAR ORDINARY DIFF. EQs

$$M_{ij} = \int_0^L \rho \, \varphi_j(x) \varphi_i(x) \, dx$$

$$A_{ij} = \int_0^L \mu \frac{\partial \varphi_j}{\partial x} \frac{\partial \varphi_i}{\partial x} \, dx$$

$\forall i,j = 1, \dots, N_h$

$$F(\varphi_i)(t) = \int_0^L f(x,t)\,\varphi_i(x)\,dx + g_L(t)\,\varphi_i(L) - g_0(t)\,\varphi_i(0)$$

$$\underline{b}(t) \longleftrightarrow \left[F(\varphi_i)\right](t)$$

$$\rightarrow \begin{bmatrix} \int_0^L f\,\varphi_0\,dx \\ \int_0^L f\,\varphi_1\,dx \\ \vdots \\ \int_0^L f\,\varphi_{N_h}\,dx \end{bmatrix} + \begin{bmatrix} \cancelto{0}{g_L(t)\,\varphi_0(L)} \\ \cancelto{0}{g_L(t)\,\varphi_1(L)} \\ \vdots \\ \underbrace{\cancelto{0}{g_L(t)\,\varphi_{N_h}(L)}}_{\substack{g_L(t) \\ (\varphi_{N_h}(L)=1)}} \end{bmatrix} - \begin{bmatrix} \overbrace{g_0(t)\,\varphi_0(0)}^{g_0(t)\quad(\varphi_0(0)=1)} \\ \cancelto{0}{\varphi_0(t)\,\varphi_1(0)} \\ \vdots \\ \cancelto{0}{g_0(t)\,\varphi_{N_h}(0)} \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} \int_0^L f\,\varphi_0\,dx \\ \vdots \\ \int_0^L f\,\varphi_{N_h}\,dx \end{bmatrix} + \begin{bmatrix} -g_0(t) \\ 0 \\ \vdots \\ 0 \\ g_L(t) \end{bmatrix}}_{\underline{b}(t)}$$

$$\boxed{M \underline{U}_1 = \frac{\Delta T^2}{2} \underline{b}_0 - \left(\frac{\Delta T^2}{2} A - M\right) \hat{\underline{U}}_0 + \Delta T M \hat{\underline{U}}_1}$$

INITIAL STEP (A)

$$M U_{k+1} = \Delta T^2 \underline{b}_k - \left(\Delta T^2 A - 2M\right) \underline{U}_k - M \underline{U}_{k-1}$$

VERIFICATION TEST :

$$u(x,t) = \cos(kx - \omega t) \qquad c = 1 \quad \mu = \rho = 1$$

- $$f(x,t) = \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2}$$

$$= -\omega^2 \cos(kx - \omega t) + k^2 \cos(kx - \omega t)$$

$$= (k^2 - \omega^2) \cos(kx - \omega t)$$

- $$g_0(t) = \mu \frac{\partial u}{\partial x}(0) = -k \sin(kx - \omega t)\Big|_{x=0}$$

$$= +k \sin(\omega t)$$

- $g_L(t) = \mu \frac{\partial u}{\partial x}(L) = -k \sin(kL - \omega t)$

- $u_0(x) = \cos(kx - \omega t)\big|_{t=0} = \cos(kx)$

- $u_1(x) = \omega \sin(kx - \omega t)\big|_{t=0} = \omega \sin(kx)$

$$k^{\cancel{2}} = \omega^2 = \cancel{k} c$$

WAVE EQUATION WITH DIFFERENT B.C.

- DIRICHLET CONDITION

$$\begin{cases} u(0,t) = g_0^D(t) \\ u(L,t) = g_L^D(t) \end{cases}$$

- ROBIN CONDITIONS $\quad a, b > 0$

$$\begin{cases} \mu \frac{\partial u}{\partial x}(0,t) - a\, u(0,t) = g_1^R(t) \\ \mu \frac{\partial u}{\partial x}(L,t) + b\, u(L,t) = g_L^R(t) \end{cases}$$

- PERIODIC CONDITIONS

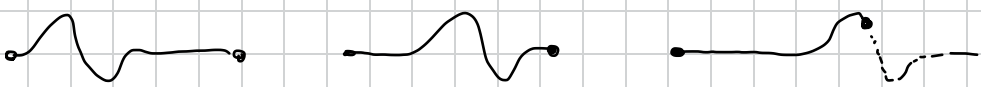
$$u(0,t) = u(L,t)$$

- IMPEDANCE CONDITIONS

$$\begin{cases} \alpha \dfrac{\partial u}{\partial t}(0,t) - c\dfrac{\partial u}{\partial x}(0,t) = g_0^I(t) \\ \alpha \dfrac{\partial u}{\partial t}(L,t) + c\dfrac{\partial u}{\partial x}(L,t) = g_L^I(t) \end{cases}$$

$$c = \sqrt{\frac{k}{\rho}} \qquad \alpha > 0$$

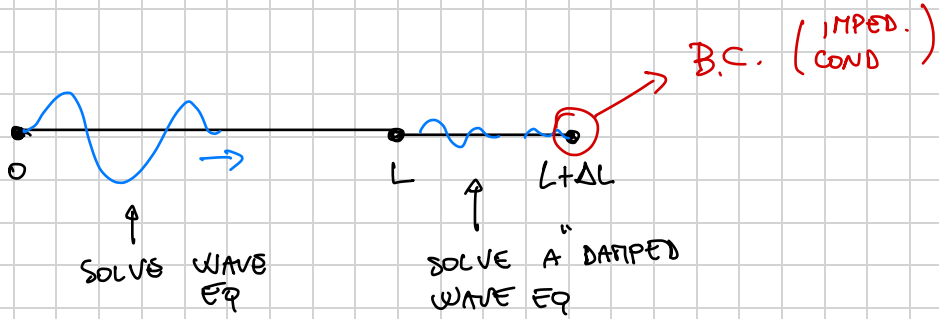
$\alpha = 1 \implies$ ABSORBING CONDITION



MIX BC.

$x=0$	$x=L$
DIR.	NEUM.
IMP.	DIR
NEUM	NEUM
$\vdots$	$\vdots$

- PERFECTLY MATCHED LAYER (PML)



1) DIRICHLET COND. $\quad u(0,t) = g_0^D(t) \qquad u(L,t) = g_L^D(t)$

$$\begin{cases} \forall t \in (0,T] \quad R^D = R^D(t) \in H^1(0,L) \\ R^D(0,t) = g_0^D(t) \\ R^D(L,t) = g_L^D(t) \end{cases}$$

$$w(x,t) = u(x,t) - R^D(x,t)$$

$$\begin{cases} \rho \dfrac{\partial^2 w}{\partial t^2} - \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial w}{\partial x}\right) = f(x,t) - \rho \dfrac{\partial^2 R^D}{\partial x^2} + \dfrac{\partial}{\partial x}\left(\mu \dfrac{\partial R^D}{\partial x}\right) \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad x \in (0,L) \quad t \in (0,T] \\ w(0,t) = w(L,t) = 0 \\ w(x,0) = u(x,0) - R^D(x,0) = u_0(x) - R^D(x,0) = \hat{u}_0(x) \\ \dfrac{\partial w}{\partial t}(x,0) = u_1(x) - \dfrac{\partial R^D}{\partial t}(x,0) = \hat{u}_1(x) \end{cases}$$

WEAK FORM: For $t \in (0,T]$ find $w(t) \in H_0^1(0,L)$

s.t.

$$\begin{cases} \displaystyle\int_0^L \rho \frac{\partial^2 w}{\partial t^2} v \, dx + \int_0^L \mu \frac{\partial w}{\partial x}\frac{\partial v}{\partial x} dx = \int_0^L f v \, dx - \int_0^L \rho R^D v \, dx - \int_0^L \mu \frac{\partial R^D}{\partial x}\frac{\partial v}{\partial x} dx \\ + \text{I.C.} \qquad \forall v \in H_0^1(0,L) \end{cases}$$

FE FORM: For $t \in (0,T]$ find $w_h \in V_{h,0}$ "V_h + ZERO ON THE BOUNDARY"

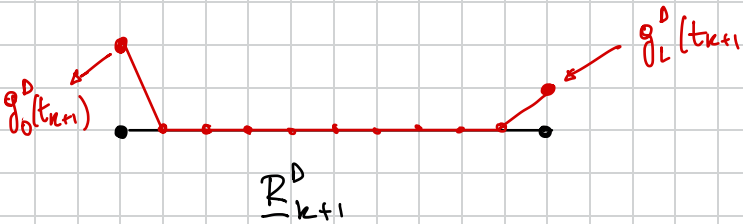
$$\begin{cases} \text{s.t.} \quad m(\ddot{w}_h, v_h) + a(w_h, v_h) = F(w_h) - m(R^D, v_h) - a(R^D, v_h) \\ + \text{I.C.} \qquad \forall v_h \in V_{h,0} \end{cases}$$

$$\begin{cases} M \ddot{\underline{w}}(t) + A \underline{w}(t) = \underbrace{\underline{b}(t)}_{\int_0^L f v \, dx} - \underbrace{M \ddot{\underline{R}}^D - A \underline{R}^D}_{\text{DIRICHLET COND}} \\ \underline{w}(0) = \hat{\underline{u}}_0 \\ \dot{\underline{w}}(0) = \hat{\underline{u}}_1 \end{cases}$$

$$M \underline{w}_{k+1} = \Delta T^2 \left(\underline{b}_k - A \underline{w}_k \right) + 2 M \underline{w}_k - M \underline{w}_{k-1}$$

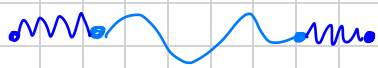
$\underline{w}_{k+1}$ that is "0" for $x=0$, $x=L$

$$\underline{U}_{k+1} = \underline{W}_{k+1} + \underline{R}^D_{k+1}$$



- ROBIN CONDITIONS $a, b > 0$

$$\begin{cases} \mu \dfrac{\partial u}{\partial x}(0,t) - a\, u(0,t) = g_0^R(t) \\ \mu \dfrac{\partial u}{\partial x}(L,t) + b\, u(L,t) = g_L^R(t) \end{cases}$$



$$\begin{cases} \mu \dfrac{\partial u}{\partial x}(0,t) = g_0^R(t) + a\, u(0,t) \\ \mu \dfrac{\partial u}{\partial x}(L,t) = g_L^R(t) - b\, u(L,t) \end{cases}$$

... IN THE WEAK FORM I HAVE

$$-\int_0^L \frac{\partial}{\partial x}\left(\mu \frac{\partial u}{\partial x}\right) v \, dx$$

$$= -\left[\mu \frac{\partial u}{\partial x} v\right]_0^L + \int_0^L \mu \frac{\partial u}{\partial x}\frac{\partial v}{\partial x}\,dx$$

$$= -\underbrace{\mu \frac{\partial u}{\partial x}(L,t)\, v(L)}_{\circledast} + \underbrace{\mu \frac{\partial u}{\partial x}(0,t)\, v(0)}_{\circledast\circledast} + \dots$$

$$= -\underbrace{g_L^R(t)\, v(L)} + \underbrace{b\, u(L,t)\, v(L)} + \underbrace{g_0^R(t)\, v(0)} + \underbrace{a\, u(0,t)\, v(0)}$$

MOVE TO THE RHS (DATA)

KEEP ON THE LHS (UNKNOWN FUNC U)

FINITE ELEMENT FORM. For any time $t \in (0,T]$ find $\underline{u}_h(t) \in V_h$ s.t.

$$\int_0^L \rho\, \ddot{u}_h(t)\, v_h\, dx + \int_0^L \mu \frac{\partial u_h}{\partial x}\frac{\partial v_h}{\partial x}\,dx$$

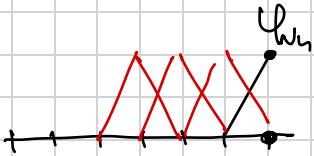
$$+ b\, u_h(L,t)\, v_h(L) + a\, u_h(0,t)\, v_h(0)$$

$$= \int_0^L f\, v_h\, dx + g_L^R(t)\, v(L) - g_0^R(t)\, v(0) \quad \forall v_h \in V_h$$

+. I.C.

$$b\, u_h(L,t)\, v_h(L) + a\, u_h(0,t)\, v_h(0) \qquad (*)$$

$$u_h(x,t) = \sum_{j=0}^{N_h} u_j(t)\, \varphi_j(x)$$



$$u_h(L,t) = \sum_{j=0}^{N_h} u_j(t)\, \underbrace{\varphi_j(L)}_{=0\ \forall j \neq N_h} = u_{N_h}(t)$$

$$u_h(0,t) = u_0(t)$$

$$v_h = \varphi_i \qquad i = 0, \dots, N_h$$

$$v_h(L) = \varphi_i(L) = 1 \iff i = N_h$$

$$v_h(0) = \varphi_i(0) = 1 \iff i = 0$$

$$(*) \quad \underbrace{\begin{pmatrix} a & 0 & \cdots & 0 \\ 0 & & & \vdots \\ \vdots & & \bigcirc & 0 \\ 0 & \cdots & \cdots & b \end{pmatrix}}_{R} \underbrace{\begin{pmatrix} u_0 \\ \vdots \\ u_{N_h} \end{pmatrix}}_{\underline{U}}$$

$$\parallel$$

$$M \ddot{\underline{U}} + (A + R)\,\underline{U} = \underline{b}(t)$$ ← I have the Robin data

COMPUTE g_0^R, g_L^R for $u(x,t) = \cos(kx - \omega t)$

$$\begin{cases} \mu \dfrac{\partial u}{\partial x}(0,t) - a\,u(0,t) = g_0^R(t) \\ \mu \dfrac{\partial u}{\partial x}(L,t) + b\,u(L,t) = g_L^R(t) \end{cases}$$

$$g_0^R(t) = -k \sin(kx - \omega t)\Big|_{x=0} - a\cos(\omega t)$$

$$= +k \sin(\omega t) - a\cos(\omega t)$$

$$g_L^R(t) = -k \sin(kL - \omega t) + b\cos(kL - \omega t)$$

$$a = b = 1$$

- PERIODIC CONDITIONS

$$\underline{u(0,t)} = \underline{u(L,t)}$$ (WEAK FORM WITH NEUMANN HOM. COND.)

$$\Downarrow$$

$$M\,\underline{U}_{k+1} = \underline{G}_{k,k-1}$$

$$\underline{U}_{k+1} = \begin{pmatrix} u_0^{k+1} \\ u_1^{k+1} \\ \vdots \\ u_{N_h}^{k+1} \end{pmatrix} \Longrightarrow u_0^{k+1} = u_{N_h}^{k+1}$$

$$\begin{array}{c}
\begin{array}{cccc} m_{11}+m_{N_h 1} & m_{12}+m_{N_h 2} & & m_{1N_h}+m_{N_h N_h} \end{array} \\
\begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1N_h} \\ m_{21} & m_{22} & \cdots & m_{2N_h} \\ \vdots & & & \\ m_{N_h 1} & m_{N_h 2} & \cdots & m_{N_h N_h} \end{pmatrix} \\
\begin{array}{ccccc} 1 & 0 & \cdots & 0 & -1 \end{array}
\end{array}
\begin{pmatrix} u_0^{k+1} \\ u_1^{k+1} \\ \vdots \\ u_{N_h}^{k+1} \end{pmatrix}
=
\begin{array}{c}
g_0^k + g_{N_h}^k \\
\begin{pmatrix} g_0^k \\ g_1^k \\ \\ g_{N_h}^k \end{pmatrix} \\
0
\end{array}$$